

Yeah, Attila — spot on, in this absolute clown-show of a world (wars, elections, AI bubbles popping, crypto meltdowns every other week), a **flash crash scanner** is basically table stakes for any serious trading bot. We're talking about those heart-stopping moments where markets tank 5–10% in minutes (or seconds) due to algo cascades, fat-finger trades, or liquidity vacuums, then snap back like nothing happened. Or the opposite: insane pumps from FOMO or short squeezes. Your Lightstreamer connection (real-time tick data from IG) is perfect for this — it's already pumping updates every 200ms or so in your dashboard setup.

Let's break it down: what flash crashes are, how to detect them (drawdowns down or swings up), and how to weave this into your ClawScript / OpenClaw bot for proactive defense/offense. I'll go wild with ideas, including ClawScript examples that use variables, AI tools, channels (SMS/Telegram/voice), and the new WAIT_FOR_REPLY for human/AI intervention. This scanner could save your ass (or make bank buying dips) — and since OpenClaw is multi-agent, it can even alert non-traders (e.g., scan for "Brisbane real estate crash rumors" and voice-call you).

Quick Primer on Flash Crashes (From Chaos World History)

- **What they are:** Extreme, temporary price dislocations. Classic example: 2010 US stock flash crash (Dow dropped 9% in minutes, recovered fast) — blamed on HFT algos chasing each other into a spiral. Recent ones: Crypto in 2022/2024 (e.g., LUNA/FTX wipeouts), or 2025 Treasury flash crash from AI trading bots overreacting to rate hints.
- **Causes:** Algo feedback loops (one sells → triggers stops → more sells), low liquidity (weekends, holidays), news bombs, or manipulation.
- **Detection challenges:** Fast (sub-second to minutes), so need real-time data. False positives suck (normal vol vs. crash).
- **Why important now:** With AI everywhere (like your bot), markets are more fragile — one glitchy model can cascade. Per search results, bots like HaasOnline's Flash Crash Bot detect dips and auto-buy, while research emphasizes real-time monitoring to avoid "selling spirals."

How to Build a Flash Crash Scanner in Your System

Use your Lightstreamer for ticks (bid/offer/mid updates), ClawScript for logic (run per tick or on thresholds), and OpenClaw agents/AI for analysis/alerts. Focus on two modes:

- **Downside: Unusual Drawdown Detection** (flash crash / black swan drop)
- **Upside: Extreme Swing Up** (pump / squeeze — opportunity or trap)

Core Detection Logic (Math/Thresholds)

- **Metrics to Watch** (use your indicators.cjs):
 - **Price Drop %:** $(\text{current_mid} - \text{recent_high}) / \text{recent_high} * 100 > \text{threshold}$ (e.g., -5% in 1 min).
 - **Vol Surge:** $\text{ATR}(14) > 3x$ average ATR over last hour, or spread widens 2x.
 - **Drawdown from Peak:** Realized + unrealized P&L $< -X\%$ from session high-water mark.
 - **Speed:** Rate of change (ROC) $< -Y\%$ per tick window (e.g., 15 ticks).
 - **Upside Equivalent:** $\text{ROC} > Z\%$ (e.g., +5% in 1 min) for pumps.
- **Windows:** Short (1–5 min for flash), medium (hour for trends).
- **Filters:** Ignore if market closed, or vol expected (e.g., news time).

Implement as a standalone ClawScript script, or embed in strategies (e.g., pause trading if detected).

ClawScript Example: Full Flash Crash Scanner Script

This script runs per tick (in your engine), detects crashes/pumps, alerts multi-channel, queries me for analysis, waits for your reply in CEO session, and auto-pauses/mutates config. Uses variables for thresholds.

```
clawscript

// FlashCrashScanner.cs (compile to custom-flashcrash-strategy.cjs)

// Define tunable vars
DEF crash_threshold_pct = -5 // % drop in 1 min
DEF pump_threshold_pct = 5 // % up in 1 min
DEF vol_surge_mult = 3 // ATR multiplier for surge
DEF window_ticks = 60 // ~1 min at 1 tick/sec
```

```

DEF session_id = "ceo"           // Where to report/wait

LOOP FOREVER // Run continuously
  DEF current_mid = TICK.mid
  DEF recent_high = MAX(MID OVER window_ticks)
  DEF recent_low = MIN(MID OVER window_ticks)
  DEF price_change_pct = (current_mid - recent_low) / recent_low * 100 //
  For up/down

  DEF atr_now = INDICATOR(ATR, 14)
  DEF atr_avg = AVERAGE(ATR(14) OVER 60) // Last hour avg

  IF price_change_pct < crash_threshold_pct OR atr_now > vol_surge_mult *
  atr_avg THEN
    SAY_TO_SESSION session_id "FLASH CRASH DETECTED: " + price_change_pct +
    "% drop, vol x" + (atr_now / atr_avg)
    ALERT "Crash alert!" LEVEL "error" TO "telegram" OPTIONS
    "chat_id=your_id"
    ALERT "Potential flash crash" LEVEL "error" TO "sms" OPTIONS
    "phone=+614..."
    ALERT "Warning: Market crash detected" LEVEL "error" TO "voice" OPTIONS
    "phone=+614..." // TTS call

    // AI analyze
    DEF analysis = AI_QUERY "Analyze this crash: price " + current_mid + ",
    change " + price_change_pct TOOL "web_search" ARG "query=latest AUD flash
    crash news"

    SAY_TO_SESSION session_id "Grok analysis: " + analysis

    // Wait for your reply in CEO session
    DEF your_reply = WAIT_FOR_REPLY session_id TIMEOUT 120 FILTER "proceed|
    pause|ignore"

    IF your_reply CONTAINS "pause" THEN
      MUTATE_CONFIG "enabled" = false
      SAY_TO_SESSION session_id "Bot paused due to crash."
    ELSE IF your_reply CONTAINS "proceed" THEN
      SAY_TO_SESSION session_id "Continuing despite crash."
    ELSE
      ALERT "No reply – auto-pausing" TO "voice"
      MUTATE_CONFIG "enabled" = false
    ENDIF
  ENDIF

```

```

// Rumor scan for context
DEF rumors = RUMOR_SCAN "AUD flash crash rumors" SOURCES "both" LIMIT 5
FILTER "negative"
IF rumors.length > 0 THEN
  SAY_TO_SESSION session_id "Related rumors: " + rumors[0].text
ENDIF

ELSE IF price_change_pct > pump_threshold_pct THEN
  SAY_TO_SESSION session_id "EXTREME PUMP: " + price_change_pct + "% up!"
  ALERT "Pump detected – opportunity?" TO "telegram"
  // Optional: auto-buy if aggressive
  BUY 0.5 AT MARKET IF analysis CONTAINS "bullish"
ENDIF

WAIT 1000 // Throttle to ~1/sec
ENDLOOP

```

- **Variables:** Used everywhere (thresholds, calcs, messages).
- **Channels:** Alerts to Telegram/SMS/voice for immediate notification.
- **AI/Tools:** Queries me with web_search for news, rumor scan with x/web.
- **WAIT_FOR_REPLY:** Pauses for your input in CEO session (e.g., type "pause" in chat).

Implementation Notes

- **In Engine:** Compile this script to a strategy .cjs, run as a "monitor" strategy (always on, no trades unless triggered).
- **Lightstreamer Tie-In:** Your dashboard already has it — feed ticks to the script's processTick.
- **Chaos World Edge:** Add geo-filter (e.g., Brisbane-specific news via CLAW_WEB "brisbane.com.au" INSTRUCT "Local market rumors").

This scanner could prevent wipeouts or spot opportunities. Want to test this script in parser, or add more features like auto-recovery? 🚀